

Algebra II with Trigonometry

Chapter 11.4

Study Guide

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Certainly! Here's a comprehensive response to your request based on the provided document.

1. Short Summary

This document serves as a study guide for "Shifted Conics" in Algebra 2 Trigonometry/Honors. It explains how conic section equations (ellipses, parabolas, hyperbolas) are shifted horizontally and vertically by altering their equations. The material details the implications of these shifts, provides identifying features for each conic (center, vertices, foci, axes lengths, asymptotes), and includes worked examples and sample problems for practice.

2. Core Definitions, Theorems, and Formulas

A. Shifting Graphs

Replacement	How the Graph is Shifted	
$x \rightarrow x - h$	Right h units	
$x \rightarrow x + h$	Left h units	
$y \rightarrow y - k$	Upward k units	
$y \rightarrow y + k$	Downward k units	

B. Standard Forms of Shifted Conics

Ellipse

- $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$
- Center: (h, k)

- Major axis: $2a$
- Minor axis: $2b$

Hyperbola

- $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ (horizontal transverse axis)
- $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$ (vertical transverse axis)
- Center: (h, k)
- Vertices: a units from center along the transverse axis
- Foci: c units from center ($c^2 = a^2 + b^2$)
- Asymptotes: Lines through center with slopes $\pm \frac{b}{a}$ or $\pm \frac{a}{b}$

Parabola

- $(x - h)^2 = 4p(y - k)$ (vertical axis, opens up/down)
- $(y - k)^2 = 4p(x - h)$ (horizontal axis, opens left/right)
- Vertex: (h, k)
- Focus: p units from vertex along axis of symmetry
- Directrix: Line p units from vertex opposite the focus

C. General Form of a Conic

- $Ax^2 + Cy^2 + Dx + Ey + F = 0$
- Parabola: If A or C is 0
- Ellipse: If A and C have the same sign ($A = C$ for a circle)
- Hyperbola: If A and C have opposite signs

3. Step-by-Step Study Guide

Step 1: Identifying Conic Sections

Use the general form $Ax^2 + Cy^2 + Dx + Ey + F = 0$:

- If A or C = 0 → Parabola
- If A and C same sign (both positive or both negative) → Ellipse (circle if A = C)
- If A and C opposite signs → Hyperbola

Step 2: Shifting the Graph

- Substitute x by $x - h$ to shift right by h
 - Substitute x by $x + h$ to shift left by h
 - Substitute y by $y - k$ to shift up by k
 - Substitute y by $y + k$ to shift down by k
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Step 3: Conic Formulas and Features

Ellipse Example

Equation: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

- Center: (h, k)
- Vertices: Move a units from center along major axis
- Foci: $c = \sqrt{a^2 - b^2}$ from center along major axis

Parabola Example

Equation: $(y - k)^2 = 4p(x - h)$

- Vertex at (h, k)
- Focus at $(h + p, k)$ (for horizontal)
- Directrix: $x = h - p$

Hyperbola Example

Equation: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$

- Center: (h, k)
 - Vertices: a units from center along x-axis
 - Foci: $c = \sqrt{a^2 + b^2}$ from center along transverse axis
 - Asymptotes: $y - k = \pm \frac{b}{a}(x - h)$
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Step 4: Find the Standard Form

- Complete the square to rewrite in shifted standard form (if needed).
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4. Practice Problems & Solutions

Problem 1

Write the equation of a parabola that opens upward, has its vertex at (2, -3), and its focus 4 units above the vertex.

Solution: Upward, so: $(x - h)^2 = 4p(y - k)$ Here $h = 2$, $k = -3$, $p = 4$
 $(x - 2)^2 = 16(y + 3)$

Problem 2

Identify the type of conic: $16x^2 - 9y^2 - 32x + 54y - 71 = 0$

Solution: Coefficients of x^2 and y^2 have opposite signs $\rightarrow$ Hyperbola.

Problem 3

Find the center and axes lengths for $\frac{(x-1)^2}{9} + \frac{(y+2)^2}{4} = 1$

Solution: Center: $(1, -2)$ Major axis: $2a = 6$ Minor axis: $2b = 4$

Problem 4

Write the standard form for an ellipse with center (0,0), vertex at (0,5), co-vertex at (3,0).

Solution: $a = 5$, $b = 3$: $\frac{y^2}{25} + \frac{x^2}{9} = 1$

Problem 5

Give the standard form of a hyperbola with center (2,-1), vertices at (2,-4) and (2,2).

Solution: Vertical transverse axis: Distance from center to vertex $a = 3$
 $\frac{(y+1)^2}{9} - \frac{(x-2)^2}{b^2} = 1$ Since vertices given, this is enough for standard form.

5. Assumptions & Ambiguities

- For finding "standard form," the assumption is the reader knows how to complete the square. Details about this process are not included in the excerpt.
 - All shifts are assumed to be standard. Any negative or positive sign errors in shifting must be double-checked.
 - The values of a , b , and c are assumed to be positive.
 - There's no explicit directrix/eccentricity formula in the excerpt, so only basic form features are used.
 - The document seems written for students already familiar with the basics of conic sections.
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End of Study Guide